An AI oracle in the style of the generic group model

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Why the generic group model and not the random oracle model

The random oracle model idealizes the *values* of a hash function: they are uniform and independent. Trained models have no such randomness to idealize. The generic group model (Shoup, EUROCRYPT 1997; Maurer, IMA 2005; for rings, Aggarwal and Maurer, EUROCRYPT 2009) idealizes *access*. Every party holds opaque handles to group elements and may only apply the group operations through the oracle. Lower bounds become provable, and they hold only for algorithms that never look at the representation. Index calculus, which does look at the representation of $\mathbb{Z}_p^*$, beats the generic bound. That is the black-box against white-box question in another setting, and the reason this note copies the generic model.

Objects

- A data distribution $\mathcal{D}$ over labelled points, and a training set S of n points.
- An architecture: a parameter space Θ and a map $\theta \mapsto f_\theta$ from parameters to functions. Write $\mathcal{F} = \{f_\theta : \theta \in \Theta\}$.
- The *fibre* of a function $f \in \mathcal{F}$ is $\Phi(f) = \{\theta \in \Theta : f_\theta = f\}$, all parameter vectors that compute it. This is the fibre of the parameterization, not of the training map.
- A training procedure Train, a randomized map from training sets to Θ. It is never opened. It is constrained only by utility.

Utility condition α**.** $\mathbb{E}_{S\sim\mathcal{D}^n}\big[\Pr_{z\sim\mathcal{D}}[f_{\mathsf{Train}(S)} \text{ correct on } z]\big] \geq \alpha$. Without it a constant output is perfectly private and nothing is learned.

The oracle

The oracle $\mathcal{O}$ keeps a private table from handles to parameter vectors and offers these operations.

Operation	Returns	Level
$\mathsf{train}(S)$	a fresh handle to $\theta \leftarrow \mathsf{Train}(S)$	call
$\mathsf{eval}(h, x)$	$f_\theta(x)$	function
$\mathsf{grad}_x(h, z)$	gradient of the loss in the input	function
$\mathsf{grad}_\theta(h, z)$	gradient of the loss in the parameters	representation
$\mathsf{reveal}(h)$	θ	representation

An *access class* is a set of permitted operations with a budget on each. Black-box is $\{\mathsf{eval}\}$. The gradient attacks of the literature use $\{\mathsf{eval}, \mathsf{grad}_\theta\}$ or $\{\mathsf{eval}, \mathsf{grad}_x\}$, and the two are different classes. Shadow-model attacks add train on chosen sets. White-box is $\{\mathsf{reveal}\}$.

The game

The challenger draws $S \sim \mathcal{D}^n$, z_1 uniformly from S, $z_0 \sim \mathcal{D}$, and b uniformly. It gives the adversary z_b and a handle to $\mathsf{Train}(S)$. The adversary uses its access class and outputs b'. Its advantage is $2\Pr[b' = b] - 1$.

Ideal representation

Assumption. Conditioned on the function $f_{\mathsf{Train}(S)}$, the parameter vector $\mathsf{Train}(S)$ is independent of S. One instance: it is uniform on the fibre $\Phi(f_{\mathsf{Train}(S)})$.

This is the idealization, in the place the random encoding holds in the generic group model. It says the weights carry nothing about the data beyond the function they compute.

Proposition. Under ideal representation, the best unbounded white-box advantage equals the best unbounded advantage of an attacker that sees only the function $f_{\mathsf{Train}(S)}$.

Proof. Under the assumption, the membership bit, the function and the parameters form a Markov chain. The best advantage of an unbounded attacker equals the total variation distance between the two conditional distributions of its view. Total variation does not increase under a channel, and the parameters are a channel applied to the function. The white-box view therefore gives at most the function's advantage, and it gives at least that much because the function is computable from the parameters. □

What the definition separates

Real training breaks the assumption, and the white-box against black-box gap splits into two parts.

1. **Representation leakage.** What the parameters say about membership beyond the function. It is zero when the assumption holds, and it is the information gap.
2. **Extraction cost.** With representation leakage zero, an efficient black-box attacker may still fail to compute from queries what the parameters hand over for free. This is the computational gap. For ReLU networks, polynomial-time extraction up to the fibre (CJM20, CCHRSS24) suggests it is small. For function families that resist learning from queries while their code reveals a secret (BGIRSVY01) it can be large.

The methodology caveat

An oracle that hands out an ideal representation of every trained function is a virtual black-box obfuscator for the trained function. BGIRSVY01 proves such obfuscators do not exist for all function families. So the ideal representation cannot be instantiated in general, as the random oracle cannot (Canetti, Goldreich, Halevi, STOC 1998). This is an analogy, recorded as heuristic, and it says the definition will have the same standing as the random oracle model: a model in which proofs are possible, with known families where it fails.

Open

- Which access classes are strictly ordered for natural $(\mathcal{D}, \Theta, \alpha)$, in particular $\{\mathsf{eval}, \mathsf{grad}_x\}$ against $\{\mathsf{eval}, \mathsf{grad}_\theta\}$.
- Lower bounds on eval and train calls for membership inference inside the model, in the way the generic group model gives a square-root lower bound for discrete logarithms.
- Whether any natural training procedure satisfies ideal representation, even approximately.